



How boundary-spanning search brings about innovation performance: the contingent effects of in house training of employees and communication investment

Jinliang Chen, Yaxin Zhao, Yu Wang & Tao Jia

To cite this article: Jinliang Chen, Yaxin Zhao, Yu Wang & Tao Jia (2024) How boundary-spanning search brings about innovation performance: the contingent effects of in house training of employees and communication investment, *Technology Analysis & Strategic Management*, 36:11, 3757-3771, DOI: [10.1080/09537325.2023.2222197](https://doi.org/10.1080/09537325.2023.2222197)

To link to this article: <https://doi.org/10.1080/09537325.2023.2222197>



Published online: 13 Jun 2023.



Submit your article to this journal [↗](#)



Article views: 337



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 3 View citing articles [↗](#)



How boundary-spanning search brings about innovation performance: the contingent effects of in house training of employees and communication investment

Jinliang Chen ^a, Yaxin Zhao^{a,b}, Yu Wang ^c and Tao Jia ^d

^aBusiness School, Central University of Finance and Economics, Beijing, People's Republic of China; ^bXiaomi Corporation, Beijing, People's Republic of China; ^cSchool of E-Business and Logistics, Beijing Technology and Business University, Beijing, People's Republic of China; ^dSchool of Management, Xi'an Jiaotong University, Xi'an, People's Republic of China

ABSTRACT

This paper investigates the impact of boundary-spanning search on innovation performance, with knowledge recombination as the underlying mechanism. It also explores the contingent effect of in house training of employees by considering not invented here syndrome derived from attitude theory, and examines communication investment as a second-order moderator. We test the hypotheses based on a sample of 184 Hong Kong manufacturing firms. The results show that boundary-spanning search is positively related to innovation performance, which is negatively moderated by in house training of employees. Moreover, communication investment has a suppressing effect on the moderating role of in house training of employees. This paper highlights the not invented here syndrome and its mechanism of attitude theory to examine the contingent effects, providing insights for managers to balance among in house training of employees, communication investment, and boundary-spanning search to enhance innovation performance.

ARTICLE HISTORY

Received 10 October 2022
Revised 26 May 2023
Accepted 31 May 2023

KEYWORDS

Not invented here syndrome; boundary-spanning search; in house training of employees; communication investment

1. Introduction

Heterogeneous and valuable knowledge is a crucial source of a firm's competitive advantage, contributing to improved innovation performance (Zhou et al. 2014). With the rise of open innovation, firms rely on external relationships to gain knowledge beyond organisational boundaries (Ferreras-Méndez et al. 2015; Yu, Yuan, and Li 2019). 'Boundary-spanning search', which acquires diverse knowledge from external sources like suppliers, customers, and competitors, has become a critical approach to this end (Laursen and Salter 2006; Wang and Guo 2019). For example, Geely, a leading Chinese automobile firm, has leveraged boundary-spanning search with its suppliers to acquire advanced technologies, enhancing market competitiveness (Wang, Li, and Chang 2016).

Boundary-spanning search has been highly advocated for knowledge acquisition (Rosenkopf and Nerkar 2001; Wang et al. 2019), as it involves multiple knowledge channels, including suppliers, customers, competitors, consultants, and universities (Laursen and Salter 2006). By leveraging boundary-spanning search, the focal firm can access not only similar but also diverse knowledge from these channels, which can enrich its internal knowledge base to improve its capacity for innovation activities (Piezunka and Dahlander 2015). Furthermore, when combined with knowledge

recombination, boundary-spanning search can facilitate incremental and radical innovations, leading to improved innovation performance (Zang et al. 2014).

Empirical research supports the idea that boundary-spanning search can promote innovation, leading to improved domain and overall performance (Nosella 2013; Rosenkopf and Nerkar 2001; Zang et al. 2014). However, prior research on the relationship between boundary-spanning search and innovation performance has not reached a consensus. In contrast with the positive effects (Laursen and Salter 2006), boundary-spanning search may hinder innovation, resulting in negative effects, due to ‘attention allocation problems’ (Koput 1997) or recombination problems (Piezunka and Dahlander 2015). To reconcile this divergence, studies have examined boundary mechanisms, considering various contingent factors like absorptive capacity, competitive intensity, and strategic R&D, to clarify the relationship’s boundary conditions (Wang and Guo 2019; Yu, Yuan, and Li 2019).

Another boundary mechanism that has gained significant attention in open innovation practices is the ‘not invented here’ (NIH) syndrome, which describes a phenomenon where an in-group rejects external knowledge (Katz and Allen 1982; Laursen and Salter 2006). The boundary conditions for the relationship between boundary-spanning search and innovation performance can be further clarified by considering NIH syndrome-related factors (Weissenberger-Eibl and Hampel 2021). Training of employees is one such factor that is closely related to NIH syndrome (Burcharth, Knudsen, and Søndergaard 2014; Westhead and Storey 1996). For instance, Hannen et al. (2019) argue that organisational training practice can reduce the negative effect of NIH syndrome. Additionally, El Maalouf and Bahemia (2022) investigated the suppressive effect of external training programmes on NIH syndrome.

External training of employees and in house training of employees are two types of training (Westhead and Storey 1996) that have been studied extensively from a human resource development perspective, aiming to enhance employees’ working competence and commitment in their firms (El Maalouf and Bahemia 2022; Pham et al. 2020). In contrast, Burcharth, Knudsen, and Søndergaard (2014) addressed the significant impact of training on changing the attitude of employees and argued that employees’ attitudes regarding open innovation can be decisive. However, there is a dearth of socio-psychology research that highlights attitude-theoretical fundamentals of constructs, especially regarding in house training of employees in adopting open innovation. Recent scholarly attention has been directed towards remedies to overcome NIH syndrome derived from in house training, such as designing a less in-group centric climate (Weissenberger-Eibl and Hampel 2021). Communication investment has been briefly mentioned in previous studies to be one such remedy, but there is limited discussion on the topic (Katz and Allen 1982; Weissenberger-Eibl and Hampel 2021). Considering these gaps, we pose the following research questions:

RQ1. What is the relationship between boundary-spanning search and innovation performance?

RQ2. How does in house training of employees influence the relationship between boundary-spanning search and innovation performance?

RQ3. How does communication investment affect the role played by in house training of employees in influencing the relationship between boundary-spanning search and innovation performance?

To address these research questions, we examine two aspects. First, based on the research on knowledge recombination, we argue the impact of boundary-spanning search on innovation performance (Wang et al. 2019). The underlying mechanism is that search depth is linked to supplementary knowledge and recombinant reuse, whereas search breadth is related to complementary knowledge and recombinant creation (Knudsen 2007). Second, NIH syndrome, which is grounded in attitude theory (Eagly and Chaiken 1993), is influenced by in house training of employees and communication investment. Due to employees’ self-concept and the resulting negative attitude towards external knowledge (Antons and Piller 2015), in house training can lead to NIH syndrome. Furthermore, communication investment can mitigate NIH syndrome through re-categorisation and affirmations (Weissenberger-Eibl and Hampel 2021).

The empirical results confirm that boundary-spanning search has a positive impact on innovation performance, which is negatively moderated by in house training of employees. Furthermore, this negative moderating effect is suppressed by communication investment. Regarding theoretical contributions, this study takes the knowledge recombination perspective to verify the positive relationship between boundary-spanning search and innovation performance. Additionally, this study investigates in house training of employees from a socio-psychological perspective to highlight its promoting effect on NIH syndrome. Finally, this study contributes to remedies for curing NIH syndrome by identifying communication investment as an effective solution.

The remainder of this paper is organised as follows. Section 2 reviews the theoretical background, followed by the development of three hypotheses in Section 3. The methodology section outlines the data collection process, construct measures, and measurement validation. Section 5 presents regression results. Finally, the paper concludes with discussions on theoretical contributions, managerial implications, and limitations and future research.

2. Theoretical background

2.1. Knowledge recombination and boundary-spanning search

Innovation, often defined as Schumpeter's creative destruction, involves improved previous or new combinations of existing and new knowledge (Schumpeter 1934). Two types of knowledge – supplementary and complementary – can be recombined to achieve innovation (Knudsen 2007; Wang et al. 2019). Regarding supplementary knowledge, companies can reuse existing recombinant patterns to efficiently meet existing customer needs, while regarding complementary knowledge, new patterns for recombination should be created to effectively meet new markets (Carnabuci and Operti 2013).

Boundary-spanning search and local search are two typical innovation search paths for firms aiming to achieve Schumpeter's creative destruction (Rosenkopf and Nerkar 2001). Unlike local search, which indicates that firms search for knowledge nearby, boundary-spanning search refers to searching for knowledge from multiple channels outside the organisational boundary (Laursen and Salter 2006; Zang et al. 2014). Search channels include, but are not limited to, suppliers, customers, competitors, and universities (Laursen and Salter 2006). Thus, boundary-spanning search enables the focal firm to acquire knowledge from different external channels that are accessible to it (Nosella 2013; Zang et al. 2014).

Boundary-spanning search can enable firms to acquire supplementary and complementary knowledge. By searching deeply in specific channels, the focal firm can become familiar with the acquired knowledge (Katila and Ahuja 2002) and obtain large quantities of supplementary knowledge (Knudsen 2007). Conversely, searching broadly across various channels, the focal firm can acquire diverse and less redundant complementary knowledge. Therefore, the effect of boundary-spanning search on innovation performance can be explained from the perspective of knowledge recombination.

2.2. Not invented here syndrome and attitude theory

Acquiring external knowledge through boundary-spanning search is undoubtedly important for innovation performance. However, boundary-spanning search may not always have positive effects. For instance, Laursen and Salter (2006) mention that there may be inadequate absorptive capacity to assimilate too much knowledge. Additionally, Katila and Ahuja (2002) argue that boundary-spanning search that is too deep may lead to rigidity. Most importantly, NIH syndrome, which indicates a phenomenon where people in groups hold negative attitudes towards external knowledge, may impede the outcomes of boundary-spanning search (Laursen and Salter 2006).

This study adopts a socio-psychological lens that highlights attitude-theoretical fundamentals of constructs (Burcharth, Knudsen, and Søndergaard 2014). In this vein, NIH syndrome indicates that in-groups reject external knowledge because of their negative attitude (Hussinger and Wastyn 2016). Katz and Allen (1982) hold that people's preference for internal knowledge, which leads to a negative attitude towards external knowledge and NIH syndrome, can be attributed to their pride in expertise and the wrong supposition of owning monopoly knowledge in their domain. Similarly, Antons and Piller (2015) clarify that internal knowledge receives more attention and is thus attached with more importance, whereas external knowledge is neglected or even rejected.

The knowledge function of attitude can also result in the filtered out of external knowledge (Antons and Piller 2015). Just as the knowledge function of attitude simplifies people's experience, the knowledge filtering mechanism of NIH syndrome simplifies the knowledge encountered by employees within the focal firm. Consequently, internal knowledge that is preferred receives more attention while external knowledge that is unfamiliar to employees is filtered out. Therefore, this knowledge filtering mechanism hampers external knowledge acquisition, leading to adverse effects on knowledge recombination (Katz and Allen 1982).

NIH syndrome is an attitude-induced bias that occurs when evaluating knowledge originating beyond organisational, spatial, or contextual boundaries (Antons and Piller 2015). Attitude, which is defined as 'a psychological tendency that is expressed by evaluating a particular entity with some degree of favor or disfavor' (Eagly and Chaiken 1993, 1), describes a person's thoughts or beliefs to influence its decisions and behaviours. According to social psychologists, attitude is a major factor influencing an individual's information processing decisions regarding external knowledge (Antons et al. 2017). Due to the negative attitude arising from perceived out-group differences, employees of the focal firm are more likely to reject external knowledge regardless of its potential value (Hannen et al. 2019).

In terms of evaluation, attitude varies in both direction and intensity. People can change their attitude when they receive messages through persuasive communication (Eagly and Chaiken 1993). Attitude object that receives more attention and better comprehension can be more persuasive, as it is weighted heavily based on information integration theory (Anderson 1971). Put differently, the weight of communication, which has a psychological impact on the individual's judgment, is a crucial determinant of employees' attitude.

2.3. In house training of employees and communication investment

Training refers to 'the systematic acquisition and development of the knowledge, skills, and attitudes required by employees to adequately perform a task or job or to improve performance in the job environment' (Tharenou, Saks, and Moore 2007, 252). Training is an important human capital investment (Almeida and Carneiro 2009; Bapna et al. 2013), which can convey task-related knowledge and skills to employees (Dokko, Wilk, and Rothbard 2009). Additionally, training is a useful tool for modifying employees' attitude (Burcharth, Knudsen, and Søndergaard 2014). Accordingly, the effect of training on organisational outcomes functions through knowledge, skills, and abilities of employees, as well as employees' attitudes that determine their behaviours (Ostroff and Bowen 2000).

Training of employees can be divided into two types: external and in house (Westhead and Storey 1996). External training of employees involves learning through external training programmes to expose employees to new ideas, while in house training refers to learning through internal education and training to equip employees with post skills. Previous studies have taken human resource development perspective to analyse the functioning of training, such as professional training, training for creativity and innovation, and open innovation training (El Maalouf and Bahemia 2022; Pham et al. 2020). However, few studies focus on training from a socio-psychology perspective, especially in house training.

Communication investment can facilitate deep communication between the focal firm's employees and external partners, promoting employees' objective comprehension of external knowledge.

Deep communication is a fundamental basis for knowledge transfer, while objective comprehension increases the likelihood of recognising, interpreting, and picking external novel knowledge (Wang and Guo 2019). Past negative attitude towards external knowledge can be changed and may evoke favourable thoughts. Therefore, communication investment can mitigate negative attitude and suppress NIH syndrome attributed to in house training.

3 Hypothesis development

3.1. *The impact of boundary-spanning search on innovation performance*

We postulate that boundary-spanning search contributes to firms' knowledge recombination, which benefits innovation performance. In terms of the depth of boundary-spanning search, the deeper a firm searches in a specific channel, the more the firm can learn from the specific partner due to the increasing amount of supplementary knowledge (Katila and Ahuja 2002). Supplementary knowledge, which is familiar to the firm, is characterised by a high level of redundancy (Knudsen 2007). By exploiting supplementary knowledge within the existing technological trajectories, the focal firm can reuse existing recombinant patterns and generate a scale economies effect, thereby contributing to innovation performance (Carnabuci and Operti 2013).

The broader a firm's boundary-spanning search is across multiple channels, the higher the degree of new recombinant patterns resulting from the diversity of complementary knowledge (Katila and Ahuja 2002). Complementary knowledge, which is less redundant, can add a greater variety of knowledge components (Carnabuci and Operti 2013). Exploring complementary knowledge enables the focal firm to create new recombinant patterns and break through existing technological trajectories, thereby promoting innovation performance. Taken together, boundary-spanning search contributes to innovation performance.

H1. A positive relationship exists between boundary-spanning search and innovation performance.

3.2. *The contingent role of in house training of employees*

We posit that in house training of employees weakens the effect of boundary-spanning search on innovation performance due to the rejection of external knowledge caused by the negative attitude of NIH syndrome. First, in house training of employees can enhance the self-concept of their organisation, which gives rise to a negative attitude towards external knowledge (Antons and Piller 2015). In house training of employees, which is used to improve employees' knowledge, skills, and abilities, can promote employees' self-concept of their organisation by increasing their commitment to the organisation's strategies and tasks (Burcharth, Knudsen, and Søndergaard 2014; El Maalouf and Bahemia 2022). Employees tend to develop a negative attitude towards external knowledge to defend their self-concept when they perceive it as a threat (Hannen et al. 2019; Hussinger and Wastyn 2016). Therefore, increasing in house training of employees can lead employees to develop an increasingly negative attitude towards external knowledge.

Furthermore, negative attitude towards external knowledge may lead to the filtering out of knowledge, which hinders knowledge recombination. NIH syndrome, described as an attitude-based bias against external knowledge, is a common phenomenon resulting from negative attitude towards knowledge, ideas, and technologies of external origin (Hussinger and Wastyn 2016). The knowledge function of attitude, which illustrates that a negative attitude leads to a biased knowledge process, underlies the filtering of external knowledge due to NIH syndrome (Antons and Piller 2015). This is because employees prefer knowledge that aligns with their attitudes, while rejecting new knowledge that challenges their attitudes. For instance, managers in charge of new product development tend to stick to their original idea regarding market potential even when contradictory evidence is presented (Antons and Piller 2015). Therefore, the relationship between boundary-

spanning search and innovation performance can be negatively influenced by in house training of employees.

H2. In house training of employees negatively moderates the relationship between boundary-spanning search and innovation performance.

3.3. The second-order contingent role of communication investment

We posit that communication investment alleviates the negative moderating effect of in house training of employees by mitigating of the negative attitude of NIH syndrome. Two mechanisms, re-categorisation and affirmations, underlie the second-order contingent role of communication investment. First, communication investment can reduce attitude bias towards external knowledge through re-categorisation. By creating a superordinate group to which employees belong and by increasing similar salience and introducing common goals between the focal firm's employees and external knowledge providers, re-categorisation can effectively reduce in-group bias (Weissenberger-Eibl and Hampel 2021). Through this mechanism, the threat to employees' self-concept is eliminated due to the change of perception 'from "us" and "them" to "we"' (Weissenberger-Eibl and Hampel 2021), and the negative attitude against external knowledge is reduced. Notably, communication investment is an effective means to facilitate deep communication between in-group employees and out-groups of external knowledge providers, which can increase similar salience and introduce common goals.

Second, communication investment can correct negative attitude towards external knowledge through affirmations. Affirmations, which are defined as 'positive communications about external knowledge providers' (Weissenberger-Eibl and Hampel 2021), can help cure NIH syndrome. Affirmations can elicit positive feelings of employees regarding external knowledge. The focal firm can establish an open innovation culture by using affirmations to transform the negative perception of 'not invented here' to 'invented anywhere,' and even to 'proudly found elsewhere', to correct the negative attitude towards external knowledge (Herzog and Leker 2010). Investment in communication can also promote positive communication between in-group employees and out-groups of external knowledge providers, generating positive cognition to mitigate negative attitude and weaken the knowledge filtering mechanism of NIH syndrome. Considering the negative moderating role of in house training of employees in influencing the relationship between boundary-spanning search and innovation performance, we predict a second-order moderating effect of communication investment.

H3. Communication investment weakens the negative moderating effect of in house training of employees on the relationship between boundary-spanning search and innovation performance.

Figure 1 presents the conceptual model.

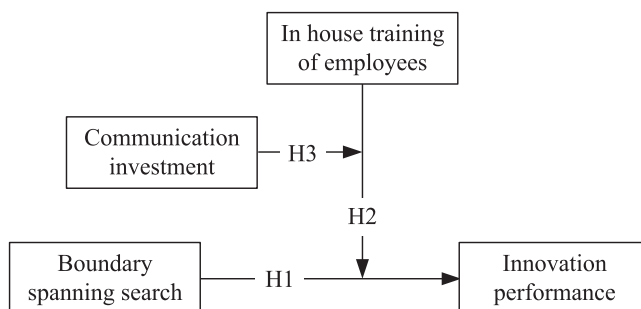


Figure 1. The conceptual model.

4. Research methodology

4.1. Sample and data collection

To test the hypotheses, a large-scale survey was conducted. The questionnaire was completed by experienced managers of Hong Kong manufacturing firms located in Dongguan city, Guangdong Province of China, who were knowledgeable about their firms' practices and performance. The survey was conducted with the support of the Dongguan municipal government and the members of the Dongguan Association of Enterprises with Foreign Investment. Additional follow-up methods such as phone calls and email contacts were utilised to increase the response rate.

We received 289 responses out of 330 sample firms, yielding a response rate of 87.6%. Ninety-five questionnaires were deleted owing to self-contradictory and blank items of focal variables, and ten questionnaires that were from non-manufacturing firms were also removed. Finally, we obtained a total of 184 valid questionnaires, resulting in an effective rate of 56%. We conducted T test between the deleted questionnaires and the remaining questionnaires to compare their firm characteristics, including total assets and number of employees. The results reveal no statistical significance, indicating that the questionnaire deletion would not impair subsequent analysis. Table 1 presents the profile of the sample.

4.2. Questionnaire design and measures

The relevant literature has been extensively studied to appropriately define the variables and measurements. As the items were based on the literature in English but the respondents were Chinese, the questionnaire was developed in English and then translated into Chinese. To ensure conceptual equivalence, the questionnaire was translated back to check for differences. Content validity was evaluated by two leaders of the association who were familiar with the sample firms and the survey objectives. Furthermore, we conducted a pilot test with 30 sample firms for minor modifications.

Independent and dependent variables

Boundary-spanning search was measured by the average communication frequency between the firm and external partners during new product or service development (Laursen and Salter 2006). Seven search channels are covered: suppliers, clients or customers, competitors, consultants, universities or other research institutes, government departments, and financial institutions. Communication frequency is considered as an effective indicator of boundary-spanning search, as it is essential for knowledge transfer (Ferrerias-Méndez et al. 2015).

Table 1. Profile of the sample.

Firm profile	Frequency	Percentage (%)	Cumulative percentage (%)
Total Assets (in million Chinese Yuan)			
0 ≤ Assets < 1	21	11.4	11.4
1 ≤ Assets < 20	102	55.4	66.8
20 ≤ Assets < 100	41	22.3	89.1
Assets ≥ 100	20	10.9	100.0
Number of Employees			
Employees < 20	3	1.63	1.63
20 ≤ Employees < 200	107	58.15	59.78
300 ≤ Employees < 1000	57	30.98	90.76
Employees ≥ 1000	17	9.24	100.0
Education Level of Employees: proportion with bachelor's degree and above (%)			
0 ≤ Education < 0.2	37	20.1	20.1
0.2 ≤ Education < 2	67	36.4	56.5
2 ≤ Education < 10	64	34.8	91.3
10 ≤ Education ≤ 100	16	8.7	100.0

Innovation performance was measured using four items adapted from Song, Dyer, and Thieme (2006). These included an item for the success of new product or service development, an item on the degree of meeting the firm's goals, and two items on excellence and short cycle with comparison to competitors. A five-point Likert scale (1 = much worse, 5 = much better) was used.

Moderating variables

In house training of employees was measured by internal education and training fees that the firm spent on employees in the last year, which were then logarithmically transformed (Almeida and Carneiro 2009; Bapna et al. 2013).

Communication investment was measured by the cost incurred in communication with external search partners. The average communication investment across the seven aforementioned search channels was calculated (Laursen and Salter 2006).

Covariates

We included five control variables: type of firm (foreign-funded or not), firm size (number of employees), total assets, education level (proportion of employees with bachelor's degree and above), and innovation intensity (ratio of independent R&D and technology introduction to annual sales), since they may affect innovation performance according to previous studies (Rubera, Chandrasekaran, and Ordanini 2016). Firm size and total assets were added into the regression after logarithm transformation.

4.3. Reliability and validity

We assessed reliability using Cronbach's alpha and conducted exploratory factor analysis to evaluate validity. Since boundary-spanning search and communication investment were measured by a single calculated item, we tested the reliability and validity of innovation performance, and the results are presented in Table 2. The Cronbach's alpha of innovation performance was above the threshold 0.70, indicating good reliability. We used maximum variance principle component analysis for exploratory factor analysis to confirm the validity of variable design. The standardised factor loading values were well above the cut-off 0.5, suggesting good aggregation validity.

5. Analysis and results

Table 3 shows the correlation matrix, mean values, and standard deviations of all variables. The mean values and standard deviations revealed no abnormalities, and the correlation coefficients were appropriate.

As shown in Table 3, all correlations are below the threshold of 0.70 (Yang and Tsai 2019). Specifically, boundary-spanning search is correlated to innovation performance ($r = 0.343$, $p < 0.001$), in house training of employees ($r = 0.310$, $p < 0.010$), and communication investment ($r = 0.586$, $p < 0.001$) in positive and significant manners. Moreover, innovation intensity is positively and significantly correlated to innovation performance ($r = 0.187$, $p < 0.050$), boundary-spanning search ($r = 0.294$, $p < 0.001$), in house training of employees ($r = 0.356$, $p < 0.001$), and communication investment ($r = 0.377$, $p < 0.001$).

Table 2. Reliability and exploratory factor analysis results.

Variables and items	Factor loading	S. E.
Innovation performance ($\alpha = 0.916$)		
Innp1: New product or service development meets company goals.	0.897	—
Innp2: New product or service development is successful.	0.936	0.056
Innp3: New products or services are better than competitors.	0.912	0.060
Innp4: New product or service cycle is shorter than competitors.	0.832	0.069

Table 3. Correlation matrix and descriptive statistics.

Variables	1	2	3	4	5	6	7	8	9
1 Innovation performance	1								
2 Boundary-spanning search	0.343***	1							
3 In house training of employees	0.103	0.310**	1						
4 Communication investment	0.237**	0.586***	0.300**	1					
5 Total assets	0.308***	0.285***	0.442***	0.122	1				
6 Firm size	0.281***	0.268***	0.515***	0.239**	-0.600***	1			
7 Education level	0.021	0.193*	0.065	0.290**	0.033	-0.087	1		
8 Firm type	0.066	0.082	0.255**	-0.003	0.348***	0.319***	0.029	1	
9 Innovation intensity	0.187*	0.294***	0.356***	0.377***	0.105	0.001	0.134	-0.100	1
Mean	3.944	2.256	0.543	1.787	3.166	2.305	0.475	0.598	2.240
SD	1.407	0.820	0.693	0.708	0.718	0.495	0.505	0.492	1.517

Note(s): $N = 184$. Significance levels are two tailed, with * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Hierarchical regression analysis was conducted to test the main effect (H1), and the results are presented in Table 4. As shown in Table 4, Model 1 contains all control variables, based on which Model 2 includes boundary-spanning search. The maximum variance inflation factor (VIF) value for the two regression models, i.e. 1.567 and 1.611, are far lower than the threshold 10, showing that multicollinearity should not a serious concern. It also shows that boundary-spanning search has a significant positive impact on innovation performance ($\beta = 0.200$, $p < 0.05$), thus supporting H1.

The SPSS version of PROCESS was used to test the moderating effect of in house training of employees (H2) and the second-order moderating effect of communication investment (H3), with results shown in Table 5. The coefficient from the two-way interaction of boundary-spanning search and in house training of employees to innovation performance is negative and significant ($\beta = -0.692$, $p < 0.01$; $CI95\% = [-1.202, -0.183]$), supporting H2. Specifically, the conditional effect of boundary-spanning search on innovation performance is positive and significant at the low level of in house training of employees ($\beta = 0.808$; $CI95\% = [0.295, 1.321]$), but not significant at the moderate level ($\beta = 0.360$; $CI95\% = [-0.024, 0.743]$), and even negative at the high level ($\beta = -0.088$; $CI95\% = [-0.586, 0.410]$). Therefore, the positive effect of boundary-spanning search on innovation performance weakens with in house training of employees.

Model 4 includes another three interactions based on Model 3 to test the second-order moderating role of communication investment. The coefficient from the three-way interaction of boundary-spanning search, in house training of employees and communication investment to innovation performance is positive and significant ($\beta = 1.606$; $CI95\% = [0.690, 2.521]$), supporting H3. In addition, the conditional moderating effect is significant at the low level of communication

Table 4. Regression results of the main effect.

	Innovation performance	
	Model 1	Model 2
Total assets	0.290*	0.243*
Firm size	0.115	0.089
Education level	-0.002	-0.049
Firm type	-0.003	0.013
Innovation intensity	0.116	0.066
Boundary-spanning search		0.243*
R ²	0.156	0.205
Adjusted R ²	0.115	0.159
F-value	3.833**	4.434***
Max VIF	1.567	1.611

Note(s): $N = 184$. The coefficients were standardised. Significance levels with * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Table 5. Regression results of the moderating effects.

	Innovation performance			
	Model 3		Model 4	
	Coeff. (SE)	[LLCI, ULCI]	Coeff. (SE)	[LLCI, ULCI]
Constant	−0.699(1.041)	[−2.776, 1.378]	−3.516(1.798)	[−7.110, 0.079]
Boundary-spanning search (BS)	0.816**(0.259)	[0.299, 1.332]	2.604*** (0.706)	[1.193, 4.015]
In house training of employees (IHTE)	1.322(0.690)	[−0.055, 2.699]	8.188*** (2.134)	[3.921, 12.456]
BS × IHTE	−0.692** (0.255)	[−1.202, −0.183]	−4.521*** (0.967)	[−6.454, −2.587]
Communication investment (CI)			1.200(1.071)	[−0.943, 3.342]
BS × CI			−0.811* (0.384)	[−1.579, −0.044]
IHTE × CI			−3.052* (1.292)	[−5.634, −0.469]
BS × IHTE × CI			1.606*** (0.458)	[0.690, 2.521]
Total assets	0.705* (0.275)	[0.156, 1.254]	0.785** (0.294)	[0.198, 1.372]
Firm size	0.442(0.383)	[−0.322, 1.206]	0.318(0.360)	[−0.403, 1.038]
Education level	−0.063(0.279)	[−0.621, 0.494]	0.003(0.292)	[−0.581, 0.587]
Firm type	−0.065(0.322)	[−0.707, 0.576]	0.042(0.319)	[−0.595, 0.680]
Innovation intensity	−0.058(0.119)	[−0.296, 0.180]	0.110(0.126)	[−0.142, 0.361]
R ² (F-value)	0.292 (3.551**)	0.437 (3.947****)		
	Coeff. (SE)	[LLCI, ULCI]	Coeff. (SE)	[LLCI, ULCI]
	In house training: Conditional direct effects of boundary-spanning search	Communication investment: Conditional moderating effects of in house training		
Mean − SD	0.808** (0.257)	[0.295, 1.321]	−2.561*** (0.520)	[−3.601, −1.520]
Mean	0.360(0.192)	[−0.024, 0.743]	−1.455*** (0.414)	[−2.284, −0.627]
Mean + SD	−0.088(0.250)	[−0.586, 0.410]	−0.350(0.521)	[−1.391, 0.691]

Note(s): $N = 184$. Significance levels with * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

investment ($\beta = -2.561$; $CI95\% = [-3.601, -1.520]$) and the moderate level ($\beta = -1.455$; $CI95\% = [-2.284, -0.627]$), but not significant at the high level ($\beta = -0.350$; $CI95\% = [-1.391, 0.691]$). That is, the negative moderating effect of in house training of employees is suppressed by increasing communication investment.

Plots of moderating effect

The interaction plots of the moderating effects of in house training of employees are presented in Figure 2, showing that the higher the in house training of employees values, the lower the slopes. That is, with the same increase in boundary-spanning search, the increase in innovation performance diminishes with in house training of employees. Figure 3 presents interaction plots of the second-order moderating effects of communication investment. Comparing the slope change from 'Low training, Low investment' to 'High training, Low investment' with the slope change from 'Low training, High investment' to 'High training, High investment', we can see that the higher communication investment values, the lower the slope change. Accordingly, the role of communication investment mitigates the negative moderating effect of in house training of employees.

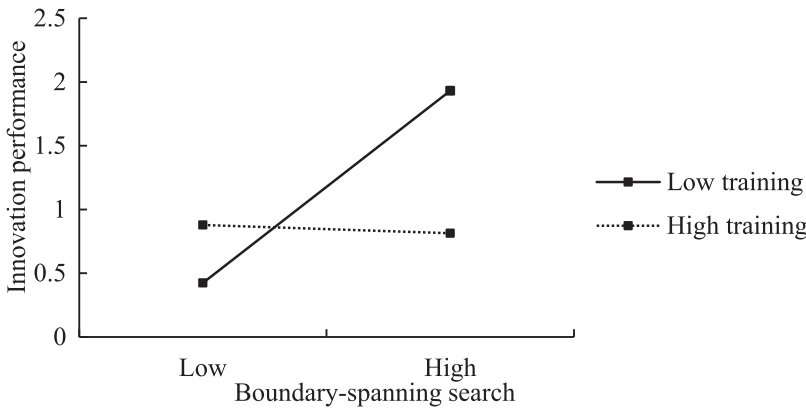


Figure 2. Plots of the moderating effects of in house training of employees.

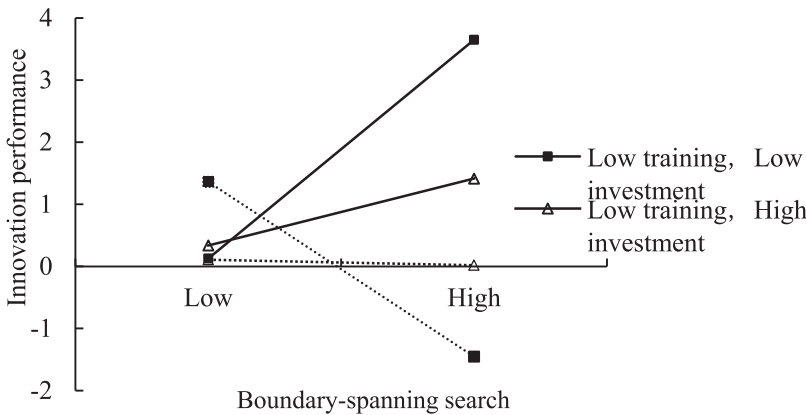


Figure 3. Plots of the second-order moderating effects of communication investment.

6. Discussion and conclusions

6.1. Research conclusion

The development of open innovation has brought attention to the relationship between boundary-spanning search and innovation performance. While most studies consider that boundary-spanning search can promote innovation performance, previous research has not reached a consensus. To reconcile this divergence, this study considered in house training of employees and communication investment as contingent factors. The results indicate that boundary-spanning search has a significant positive impact on innovation performance due to the acquisition of supplementary and complementary knowledge. Additionally, in house training of employees significantly and negatively moderates the relationship between boundary-spanning search and innovation performance, which increases employees' self-concept and then negative attitude towards external knowledge, namely NIH syndrome. Communication investment has a significant and positive moderating role in mitigating the negative influence of in house training of employees by promoting recategorisation and affirmations, which helps suppress NIH syndrome.

6.2. Theoretical contributions

The theoretical contributions of this study are as follows. First, this study confirms the knowledge recombination mechanism underlying the positive effect of boundary-spanning search on innovation performance. Specifically, we analyse that search depth is related to supplementary knowledge and recombinant reuse, while search breadth is related to complementary knowledge and recombinant creation, which promotes innovation performance by improving existing products and developing new ones. This finding is in line with previous studies that also found a positive relationship between boundary-spanning search and innovation performance (Ferreras-Méndez et al. 2015).

Second, by introducing the attitude theory underlying NIH syndrome, this study expands the analysis perspective on the negative moderating role of training in the relationship between boundary-spanning search and innovation performance. NIH syndrome, a phenomenon in which people reject external knowledge, can be explained based on attitude theory. Previous studies have clarified the suppressive effects of various forms of training, such as professional training, training for creativity and innovation, and open innovation training, on NIH syndrome (Burcharth, Knudsen, and Søndergaard 2014; El Maalouf and Bahemia 2022). Drawing on the socio-psychology perspective, this study extends the training's negative moderating role by highlighting the promoting effect of in house training on NIH syndrome due to employees' self-concept to their organisations.

Third, this study enriches the remedies to overcome NIH syndrome by demonstrating the weakening effect of communication investment on the moderating role of in house training of employees. Communication investment reflects the efforts made by the focal firm to establish deep communication with its external partners and enhance objective comprehension of external knowledge, which mitigates negative attitude of NIH syndrome due to re-categorisation and affirmations. This finding contributes to existing countermeasures, such as reducing in-group design (Burcharth and Fosfuri 2015), promoting cross-project teams (Kathoefer and Leker 2012), and fostering an open climate for innovation (Herzog and Leker 2010).

6.3. Managerial implications

First, firms should conduct extensive knowledge search to achieve higher innovation performance. Managers should aim for a boundary-spanning search, which involves deep and broad search of external knowledge from multiple channels, including suppliers, customers, competitors,

universities, and others to acquire supplementary and complementary knowledge to enrich their knowledge base. Furthermore, they should provide management support and resources for recombinant reuse and creation to facilitate efficient and effective knowledge recombination. Therefore, boundary-spanning search can serve as a reliable option for the focal firms seeking to improve their innovation performance.

Second, managers should be aware of the potential negative attitude employees may have towards external knowledge, as it can hinder the acquisition and utilisation of external knowledge. While in house training of employees is widely acknowledged as a learning process for employees to acquire specific knowledge and technologies, it can also cultivate an organisational self-concept among their employees and limit their openness to external knowledge, leading to the development of negative attitude towards it. The negative attitude should be a concern for managers, and they should strive to balance in house training of employees with boundary-spanning search to improve innovation performance.

Third, increasing communication investment can help focal firms to suppress the impact of negative attitude due to NIH syndrome caused by in house training of employees. When managers devote in house training to equip their employees with specific knowledge and technologies, they should also allocate resources to communication efforts to address the induced negative attitude towards external knowledge. By increasing communication investment, managers can change employees' negative attitude and weaken knowledge filtering mechanisms. Thus, increasing communication investment is an effective way to reconcile potential conflict between boundary-spanning search and in house training of employees.

6.4. Limitations and future research

Considering the limitations of this study, future research can benefit from the following directions. First, while we studied boundary-spanning search by examining the overall effect of multiple channels on innovation performance, future research can examine their individual effects for more specific findings. Second, we explored NIH syndrome as a potential mechanism based on attitude theory. Future research can measure it with a structured scale to directly examine its role. Third, this study conducted quantitative research to examine the moderating role played by in house training, future research can rely on in-depth qualitative analysis to assess whether and how the effects of in house training of employees depend on the content of the cursus. Finally, this study utilised cross-sectional data to test hypotheses, which is limited in testing causality. For instance, total assets are correlated to boundary-spanning search, in house training of employees, and communication investment, which indicates that spending in communication and training may just be a consequence of innovation performance. To address these issues, future research can collect longitudinal data to test the causal inference while examining the evolution of search patterns to better guide firms' search practices.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This work was supported by National Natural Science Foundation of China: [Grant Number 71302128]; Beijing Social Science: [Grant Number 22GLC044]; National Office for Philosophy and Social Sciences: [Grant Number 19BGL096]; The Research Foundation for Youth Scholars of Beijing Technology and Business University; Yunnan Deng Ronglin Education Foundation; The 2021 Undergraduate Teaching Reform Innovation Project of Higher Education in Beijing; The 2022 Education and Teaching Reform Project of Central University of Finance and Economics [Grant Number 2022ZXJG16]; The Program for Innovation Research in Central University of Finance and Economics; Innovation Centre for Digital Business and Capital Development of Beijing Technology and Business University [Grant Number SZSK202316].

Notes on contributors

Jinliang Chen is currently an Associate Professor at Business School, Central University of Finance and Economics. He received the PhD degree in Business Management from Renmin University of China in 2010. His research interests include supply chain management and innovation management. He has co-authored articles in *International Journal of Operations and Production Management*, etc.

Yaxin Zhao received her master degree from the Business School, Central University of Finance and Economics. She is now working in Xiaomi corporation. Her research interest is innovation management. She has co-authored articles in *Foreign Economics & Management* (in Chinese), etc.

Yu Wang (corresponding author) is currently a Lecturer at School of E-Business and Logistics, Beijing Technology and Business University. She received her PhD degrees from Xi'an Jiaotong University and City University of Hong Kong in 2021. Her research interests include supply chain innovation management. She has co-authored articles in *Supply Chain Management: An International Journal*, etc.

Tao Jia is currently a Professor at School of Management, Xi'an Jiaotong University. He received the PhD degree from Xi'an Jiaotong University in 2005. His research interests comprise production and operation management. He has co-authored articles in *European Journal of Operational Research*, etc.

ORCID

Jinliang Chen  <http://orcid.org/0000-0002-6555-3584>

Yu Wang  <http://orcid.org/0000-0002-3705-4542>

Tao Jia  <http://orcid.org/0000-0002-0256-4315>

References

- Almeida, R., and P. Carneiro. 2009. "The Return to Firm Investments in Human Capital." *Labour Economics* 16 (1): 97–106. doi:10.1016/j.labeco.2008.06.002.
- Anderson, N. H. 1971. "Integration Theory and Attitude Change." *Psychological Review* 78 (3): 171–206. doi:10.1037/h0030834.
- Antons, D., M. Declerck, K. Diener, I. Koch, and F. T. Piller. 2017. "Assessing the not-Invented-Here Syndrome: Development and Validation of Implicit and Explicit Measurements." *Journal of Organizational Behavior* 38 (8): 1227–1245. doi:10.1002/job.2199.
- Antons, D., and F. T. Piller. 2015. "Opening the Black Box of "Not Invented Here": Attitudes, Decision Biases, and Behavioral Consequences." *Academy of Management Perspectives* 29 (2): 193–217. doi:10.5465/amp.2013.0091.
- Bapna, R., N. Langer, A. Mehra, R. Gopal, and A. Gupta. 2013. "Human Capital Investments and Employee Performance: An Analysis of it Services Industry." *Management Science* 59 (3): 641–658. doi:10.1287/mnsc.1120.1586.
- Burcharth, A. L. D. A., and A. Fosfuri. 2015. "Not Invented Here: How Institutionalized Socialization Practices Affect the Formation of Negative Attitudes Toward External Knowledge." *Industrial and Corporate Change* 24 (2): 281–305. doi:10.1093/icc/dtu018.
- Burcharth, A. L. D. A., M. P. Knudsen, and H. A. Søndergaard. 2014. "Neither Invented nor Shared Here: The Impact and Management of Attitudes for the Adoption of Open Innovation Practices." *Technovation* 34 (3): 149–161. doi:10.1016/j.technovation.2013.11.007.
- Carnabuci, G., and E. Operti. 2013. "Where do Firms' Recombinant Capabilities Come from? Intraorganizational Networks, Knowledge, and Firms' Ability to Innovate Through Technological Recombination." *Strategic Management Journal* 34 (13): 1591–1613. doi:10.1002/smj.2084.
- Dokko, G., S. L. Wilk, and N. P. Rothbard. 2009. "Unpacking Prior Experience: How Career History Affects job Performance." *Organization Science* 20 (1): 51–68. doi:10.1287/orsc.1080.0357.
- Eagly, A. H., and S. Chaiken. 1993. *The Psychology of Attitudes*, 1–4. Belmont, CA: Wadsworth, Thomson.
- El Maalouf, N., and H. Bahemia. 2022. "The Implementation of Inbound Open Innovation at the Firm Level: A Dynamic Capability Perspective." *Technovation* 122: 102659. doi:10.1016/j.technovation.2022.102659.
- Ferreras-Méndez, J. L., S. Newell, A. Fernández-Mesa, and J. Alegre. 2015. "Depth and Breadth of External Knowledge Search and Performance: The Mediating Role of Absorptive Capacity." *Industrial Marketing Management* 47: 86–97. doi:10.1016/j.indmarman.2015.02.038.
- Hannen, J., D. Antons, F. Piller, T. O. Salge, T. Coltman, and T. M. Devinney. 2019. "Containing the not-Invented-Here Syndrome in External Knowledge Absorption and Open Innovation: The Role of Indirect Countermeasures." *Research Policy* 48 (9): 103822. doi:10.1016/j.respol.2019.103822.
- Herzog, P., and J. Leker. 2010. "Open and Closed Innovation: Different Innovation Cultures for Different Strategies." *International Journal of Technology Management* 52 (3/4): 322–343. doi:10.1504/IJTM.2010.035979.

- Hussinger, K., and A. Wastyn. 2016. "The NIH Syndrome, Knowledge Sources and Firm Success." *R&D Management* 46 (S3): 945–957. doi:[10.1111/radm.12136](https://doi.org/10.1111/radm.12136).
- Kathoefer, D. G., and J. Leker. 2012. "Knowledge Transfer in Academia: An Exploratory Study on the not-Invented-Here Syndrome." *The Journal of Technology Transfer* 37 (5): 658–675. doi:[10.1007/s10961-010-9204-5](https://doi.org/10.1007/s10961-010-9204-5).
- Katila, R., and G. Ahuja. 2002. "Something Old, Something New: A Longitudinal Study of Search Behavior and New Product Introduction." *Academy of Management Journal* 45 (6): 1183–1194. doi:[10.2307/3069433](https://doi.org/10.2307/3069433).
- Katz, R., and T. J. Allen. 1982. "Investigating the Not Invented Here (NIH) Syndrome: A Look at the Performance, Tenure, and Communication Patterns of 50 R&D Project GROUPS." *R&D Management* 12 (1): 7–20. doi:[10.1111/j.1467-9310.1982.tb00478.x](https://doi.org/10.1111/j.1467-9310.1982.tb00478.x).
- Knudsen, M. P. 2007. "The Relative Importance of Interfirm Relationships and Knowledge Transfer for New Product Development Success." *Journal of Product Innovation Management* 24 (2): 117–138. doi:[10.1111/j.1540-5885.2007.00238.x](https://doi.org/10.1111/j.1540-5885.2007.00238.x).
- Koput, K. W. 1997. "A Chaotic Model of Innovative Search: Some Answers, Many Questions." *Organization Science* 8 (5): 528–542. doi:[10.1287/orsc.8.5.528](https://doi.org/10.1287/orsc.8.5.528).
- Laursen, K., and A. Salter. 2006. "Open for Innovation: The Role of Openness in Explaining Innovation Performance among UK Manufacturing Firms." *Strategic Management Journal* 27 (2): 131–150. doi:[10.1002/smj.507](https://doi.org/10.1002/smj.507).
- Nosella, A. 2013. "Search Practices in the Early Phase of the Innovation Process and Ambidexterity: Testing a Sample of High-Tech Companies." *Technology Analysis & Strategic Management* 26 (2): 135–153. doi:[10.1080/09537325.2013.850156](https://doi.org/10.1080/09537325.2013.850156).
- Ostroff, C., and D. E. Bowen. 2000. "Moving HR to a Higher Level: Hr practices and Organizational Effectiveness." In *Multilevel Theory, Research, and Methods in Organizations*, edited by K. J. Klein, and S. W. Kozlowski, 211–266. San Francisco, CA: Jossey-Bass.
- Pham, N. T., T. Vo Thanh, Z. Tučková, and V. T. N. Thuy. 2020. "The Role of Green Human Resource Management in Driving Hotel's Environmental Performance: Interaction and Mediation Analysis." *International Journal of Hospitality Management* 88 (1): 102392. doi:[10.1016/j.ijhm.2019.102392](https://doi.org/10.1016/j.ijhm.2019.102392).
- Piezunka, H., and L. Dahlander. 2015. "Distant Search, Narrow Attention: How Crowding Alters Organizations' Filtering of Suggestions in Crowdsourcing." *Academy of Management Journal* 58 (3): 856–880. doi:[10.5465/amj.2012.0458](https://doi.org/10.5465/amj.2012.0458).
- Rosenkopf, L., and A. Nerkar. 2001. "Beyond Local Search: Boundary-Spanning, Exploration, and Impact in the Optical Disk Industry." *Strategic Management Journal* 22 (4): 287–306. doi:[10.1002/smj.160](https://doi.org/10.1002/smj.160).
- Rubera, G., D. Chandrasekaran, and A. Ordanini. 2016. "Open Innovation, Product Portfolio Innovativeness and Firm Performance: The Dual Role of new Product Development Capabilities." *Journal of the Academy of Marketing Science* 44 (2): 166–184. doi:[10.1007/s11747-014-0423-4](https://doi.org/10.1007/s11747-014-0423-4).
- Schumpeter, J. A. 1934. *The Theory of Economic Development: An Inquiry Into Profits, Capital, Credit, Interest, and the Business Cycle*. Cambridge, MA: Harvard University Press.
- Song, M., B. Dyer, and R. J. Thieme. 2006. "Conflict Management and Innovation Performance: An Integrated Contingency Perspective." *Journal of the Academy of Marketing Science* 34 (3): 341–356. doi:[10.1177/0092070306286705](https://doi.org/10.1177/0092070306286705).
- Tharenou, P., A. M. Saks, and C. Moore. 2007. "A Review and Critique of Research on Training and Organizational-Level Outcomes." *Human Resource Management Review* 17 (3): 251–273. doi:[10.1016/j.hrmr.2007.07.004](https://doi.org/10.1016/j.hrmr.2007.07.004).
- Wang, Y., and B. Guo. 2019. "Managing External Knowledge Search: The Multiple and Contingent Roles of Absorptive Capacity." *Technology Analysis & Strategic Management* 32 (1): 29–43. doi:[10.1080/09537325.2019.1632821](https://doi.org/10.1080/09537325.2019.1632821).
- Wang, Y., T. Jia, J. Chen, and H. Sun. 2019. "Recombine supplier-side search via innovation ambidexterity: an empirical study on Hong Kong manufacturing firms." *International Journal of Physical Distribution & Logistics Management* 49 (2): 178–199. doi:[10.1108/IJPDLM-02-2018-0054](https://doi.org/10.1108/IJPDLM-02-2018-0054).
- Wang, J. J., J. J. Li, and J. Chang. 2016. "Product Co-Development in an Emerging Market: The Role of Buyer-Supplier Compatibility and Institutional Environment." *Journal of Operations Management* 46 (1): 69–83. doi:[10.1016/j.jom.2016.07.002](https://doi.org/10.1016/j.jom.2016.07.002).
- Weissenberger-Eibl, M. A., and T. I. M. Hampel. 2021. "What do we Have in-Common? Overcoming the not-Invented-Here Syndrome Through Recategorisation." *International Journal of Innovation Management* 25 (6): 2150070. doi:[10.1142/S1363919621500705](https://doi.org/10.1142/S1363919621500705).
- Westhead, P., and D. Storey. 1996. "Management Training and Small Firm Performance: Why is the Link so Weak?" *International Small Business Journal* 14: 13–24. doi:[10.1177/0266242696144001](https://doi.org/10.1177/0266242696144001).
- Yang, S. Y., and K. H. Tsai. 2019. "Lifting the Veil on the Link Between Absorptive Capacity and Innovation: The Roles of Cross-Functional Integration and Customer Orientation." *Industrial Marketing Management* 82 (1): 117–130. doi:[10.1016/j.indmarman.2019.02.006](https://doi.org/10.1016/j.indmarman.2019.02.006).
- Yu, Y., L. Yuan, and J. Li. 2019. "Knowledge Search Modes and Innovation Performance: The Moderating Role of Strategic R&D Orientation." *Technology Analysis & Strategic Management* 31 (6): 625–640. doi:[10.1080/09537325.2018.1541172](https://doi.org/10.1080/09537325.2018.1541172).
- Zang, J., C. Zhang, P. Yang, and Y. Li. 2014. "How Open Search Strategies Align with Firms' Radical and Incremental Innovation: Evidence from China." *Technology Analysis & Strategic Management* 26 (7): 781–795. doi:[10.1080/09537325.2014.899345](https://doi.org/10.1080/09537325.2014.899345).
- Zhou, K. Z., Q. Zhang, S. Sheng, E. Xie, and Y. Bao. 2014. "Are Relational Ties Always Good for Knowledge Acquisition? Buyer-Supplier Exchanges in China." *Journal of Operations Management* 32 (3): 88–98. doi:[10.1016/j.jom.2014.01.001](https://doi.org/10.1016/j.jom.2014.01.001).